MD MAHMUD AKON

Power Systems & Microgrids | Power Electronics & Control | AI for Energy | Renewable Energy & Storage
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Research Profile

Electrical and Electronic Engineering graduate from the Bangladesh University of Engineering and Technology (BUET), specializing in Electrical Energy and Power Systems. Research experience spans grid-forming inverter control, PV–battery microgrids, renewable charging systems, battery modeling, grid-data forecasting, and physics-guided machine learning. Interested in combining power-system and converter modeling, control, optimization, and data-driven methods to improve the stability, reliability, resilience, and sustainability of future energy systems.

Education

Bangladesh University of Engineering and Technology (BUET) *Jan 2022 – Jun 2026*
B.Sc. Engg. in Electrical and Electronic Engineering
Major: Electrical Energy and Power Systems (EEPS)
CGPA: 3.16/4.00 **(3.41/4.00, final two semesters)**
Coursework: Power System I & II; Power Electronics; Control Systems I; Smart Grid; Power System Operation and Control; Power Transmission and Distribution; Power System Protection; High Voltage Engineering; Linear Algebra; Numerical Technique Laboratory

Notre Dame College *2020*
Higher Secondary Certificate (HSC), Dhaka, Bangladesh
GPA: 5.00/5.00 | Ranked 61st of 292,160, Dhaka Board
Talentpool Merit Scholarship, Government of Bangladesh

Mahilara A. N. Secondary School *2018*
Secondary School Certificate (SSC), Gournadi, Barisal, Bangladesh
GPA: 5.00/5.00 | Ranked 1st of 1,500+, Gournadi Upazila, Barisal Board
Talentpool Merit Scholarship, Government of Bangladesh

Research Interests

- **Power systems and microgrids:** Planning, operation, energy management, distributed energy resources, reliability, resilience, and power quality.
- **Power electronics and converter control:** Grid-forming inverters, virtual synchronous generators, renewable-energy conversion, and operation under weak-grid and converter constraints.
- **AI and optimization for energy systems:** Controller tuning, safe reinforcement learning, forecasting, uncertainty-aware optimization, and interpretable physics-guided learning.
- **Renewable energy, storage, and electric mobility:** PV–battery systems, grid-aware EV charging, battery modeling, renewable forecasting, and storage coordination.

Technical Skills

- **Programming and Scientific Computing:** Python, MATLAB, C, C++
- **Power-System Simulation and Analysis:** OpenDSS, Siemens PSS®E, CYME PSAF, MATLAB/Simulink
- **Power Electronics and Hardware Design:** LTspice, PSpice, EasyEDA, ESP32, STM32, Arduino
- **Battery and Physics-Based Modeling:** PyBaMM, DFN/P2D modeling, electrochemical simulation
- **Machine Learning:** Supervised and unsupervised learning, reinforcement learning, time-series forecasting, physics-guided learning
- **Software and Development Tools:** GitHub, Firebase, Google Cloud, HTML, CSS, JavaScript

Publications and Submitted Manuscripts

1. **Interpretable Physics-Guided Region-Aware Attention Learning for Series and Shunt Resistance Estimation from Photovoltaic I–V Characteristics.**
   Second author | Published, Engineering Research Express (IOP Publishing), 2026
   DOI: 10.1088/2631-8695/aead4c

   Contributed to a physics-guided learning framework for estimating photovoltaic series and shunt resistance from I–V characteristics, combining region-aware attention and physical constraints for interpretable parameter estimation.

2. **SOC- and Current-Constrained VSG Supervision for Grid-Forming PV–BESS Microgrids.**
   First author | Submitted to the 14th International Conference on Electrical and Computer Engineering (ICECE 2026)

   Proposes a hybrid VSG supervisor that coordinates inertia and damping with battery SOC, energy reserves, and converter-current limits. Evaluates constraint coordination using equal-budget controller comparisons and randomized reduced-order simulations.

3. **Dual-Layer Renewable Charging for Electric Rickshaws in Bangladesh: Vehicle-Integrated PV and Grid-Aware Solar-Wind-Battery Management.**
   First author | Submitted to the 14th International Conference on Electrical and Computer Engineering (ICECE 2026)

   Models vehicle-integrated PV and a solar–wind–battery charging station with grid-import limits and priority scheduling. Simulations across three Bangladeshi locations assess charging service, grid demand, station sizing, and wind-resource viability.

Research Experience

Trustworthy AI-Based Tuning and Safe Adaptation of Grid-Forming Inverters
Ongoing research
Supervisor: Shuvangkar Das, Senior Research Engineer, INS Engineering.

- Investigating uncertainty-aware, multi-objective auto-tuning of grid-forming controllers under varying grid strength.
- Developing safe residual reinforcement-learning approaches for adaptive virtual damping and impedance control in weak grids.

Convergence Basin: Initialization Sensitivity of DRL-Tuned Inverter Current-Controller Gains
Ongoing research
Supervisor: Shuvangkar Das, Senior Research Engineer, INS Engineering.

- Investigating the sensitivity of deep reinforcement learning-based inverter current-controller tuning to initialization.
- Analyzing relationships among initialization, convergence behavior, and controller performance.

Coupled Electrochemical–Mechanical Modelling and Sensitivity-Based Optimization of Silicon–Graphite Composite Lithium-Ion Battery Anodes
Undergraduate Thesis, Defended Jun 2026
Supervisor: Dr. Md. Zahurul Islam, Professor, EEE, BUET; Visiting Professor, University of Alberta.

- Developed a coupled electrochemical–mechanical model in PyBaMM using the DFN/P2D framework to analyze capacity, stress evolution, crack-growth risk, and degradation in silicon–graphite composite anodes.
- Performed sensitivity-based analysis to study the influence of material and operating parameters on electrochemical and mechanical performance.

Selected Research and Engineering Projects

PGCB Power Data Visualizer *Jun 2026*
Developed an open-source platform for automated collection, visualization, and analysis of Bangladesh national-grid data. Implemented 24-hour machine-learning-based demand and fuel-mix forecasting using historical and real-time data. Project Link

Performance Analysis of HVDC Integration and Harmonic Mitigation in an IEEE 14-Bus System
Aug 2025
Modeled HVDC integration in the IEEE 14-bus system using PSS®E; evaluated harmonics, network performance, and mitigation strategies.

Solar Charge Controller with Maximum Power Point Tracking *Dec 2024*
Designed an MPPT-based solar charge controller to improve photovoltaic energy extraction and battery-charging efficiency using power-electronic control techniques.

IoT-Based Smart Energy Meter *Feb 2024*
Developed an IoT-enabled energy-monitoring system for real-time measurement, remote acquisition, and visualization of electrical consumption data.

Industrial Training

Industrial Attachment, Power Grid Bangladesh PLC (PGCB) *16–23 Nov 2025*

- Completed a six-day attachment covering transmission planning, grid operation, protection, and asset diagnostics.
- Gained exposure to 400/230/132 kV substations, PSS®E studies, and SCADA/NLDC operation.
- Observed relay coordination and transformer insulation and oil testing.

Professional Experience

Research Associate and Physics Instructor *Mar 2023 – Present*
Utkorsho, OnnoRokom Group
Develop structured STEM learning resources and research-oriented content; teach physics with emphasis on scientific reasoning and problem-solving. Since Nov 2025, also contribute to R&D and product development, including technical planning and workflow improvement for learner-centered digital platforms.

Physics Instructor *May 2023 – Present*
Udvash
Teach physics for board and competitive examinations and develop problem-solving-based academic materials.

Leadership and Service

Programme Committee Chair *Jun 2025 – Jun 2026*
IEEE BUET Student Branch
Led the planning and execution of technical, professional-development, and student-engagement programs.

Webmaster *Jan 2024 – Jun 2025*
IEEE BUET Student Branch
Managed the branch website, digital communications, and technical updates.

Technical Activities Coordinator *Sep 2023 – Nov 2025*
IEEE Electron Devices Society BUET Student Branch Chapter
Supported technical workshops, seminars, and knowledge-sharing activities.

Student Representative *Three Years*
Represented BUET Batch 20 and served four terms as Class Representative, coordinating communication between students and academic administration.

References

Dr. Md. Zahurul Islam
Thesis Supervisor
Professor, EEE, BUET
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Dr. Lutfa Akter
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Shuvangkar Das
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Senior Research Engineer
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